Keyword	Definition
Force	A push or pull that can change the motion, direction, or shape of an object.
Newton's First Law	An object stays at rest or moves at constant speed in a straight line unless a resultant force acts on it.
Newton's Second Law	The acceleration of an object is proportional to the resultant force (Force = Mass × Acceleration).
Newton's Third Law	If an object A applies a force on object B, object B will apply a force on object A with the same magnitude but opposite direction.
Mass (kg)	The amount of matter in an object. It stays the same everywhere in the universe.
Weight (N)	The gravitational force acting on a mass. Weight = Mass × Gravitational Field Strength.
Resultant Force	The single overall force found by combining all component forces acting on an object.
Balanced Forces	Forces that are equal in size and opposite in direction, so the object's motion does not change.
Unbalanced Forces	Forces that are not equal, causing acceleration, deceleration, or a change in direction.
Acceleration, a	When an object speeds up. It happens when a net (unbalanced) force acts on it.
Deceleration	When an object slows down because a force acts in the opposite direction to its motion.
Momentum, p	The product of mass and velocity. (Momentum = Mass × Velocity) The unit is kg.m/s
Impulse, J	The change in momentum caused by a force acting over time. Impulse = Force × Time.
F–t Graph (Force–Time Graph)	A graph showing how force changes with time. The area under the graph represents impulse (change in momentum).
Principle of the Conservation of Momentum	The total momentum before a collision is equal to the total momentum after, if no external forces act.
Scalars	Quantities that have only size (magnitude), such as speed or distance.
Vectors	Quantities that have both size and direction, such as velocity or force.
Velocity	Speed in a particular direction.
Speed	How fast an object is moving (distance divided by time).
Deform	When a shape changes because a force is applied, such as stretching, squashing, or bending.
Impulse–Momentum Relationship	States that Force × Time = Change in Momentum.
Reaction Force	The force that acts in the opposite direction when one object exerts a force on another.
Resultant Vector	The overall vector found when two or more vectors are combined (added or subtracted).
Component Forces (vertical or horizontal)	Forces that act along the horizontal and vertical directions and combine to form a resultant force.